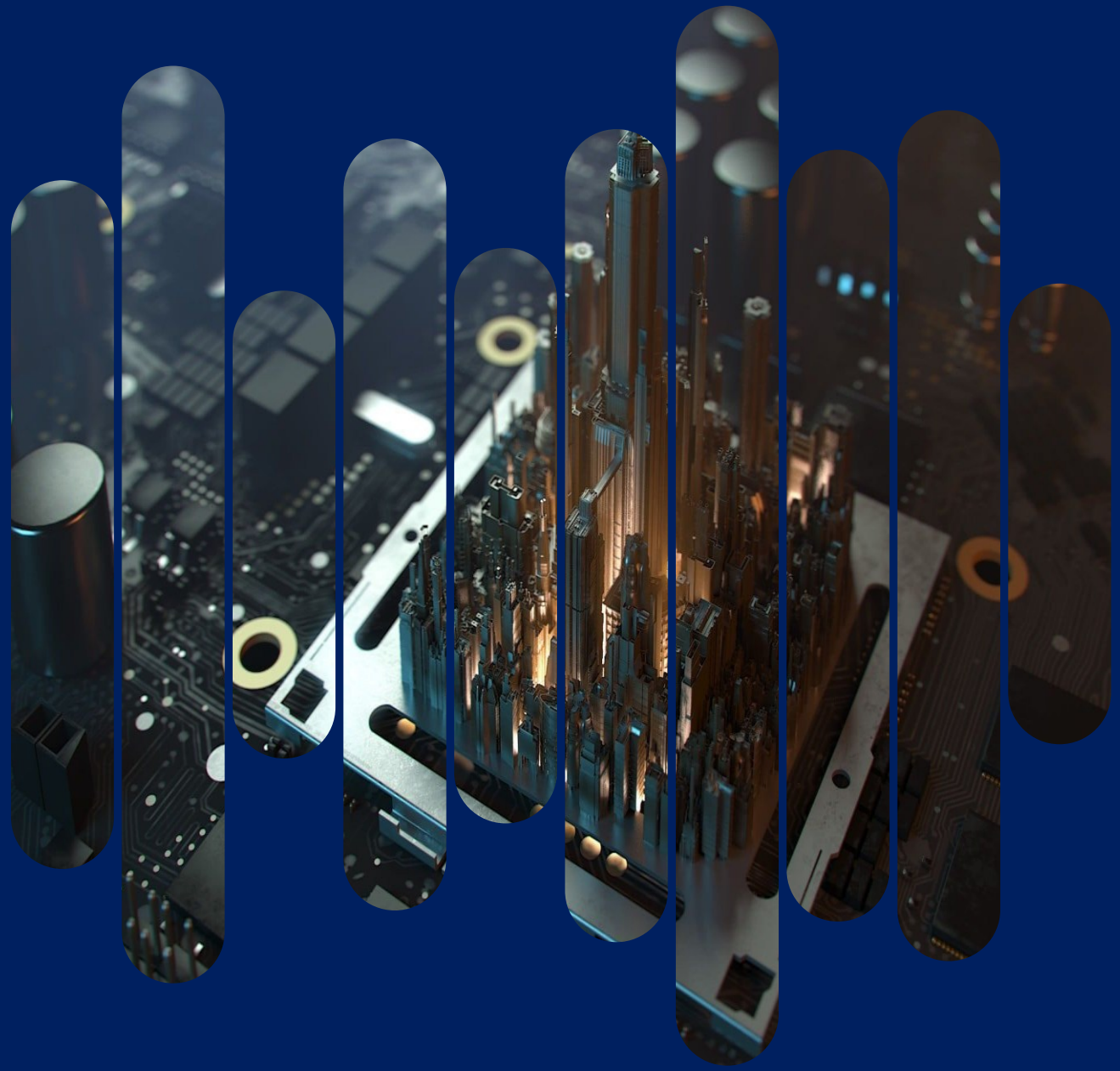
Microprocessor and Computer Architecture



CHAPTER 1

An overview of computer and Microprocessor

1.1 Brief overview of microprocessor and microcontroller

1.2 Organization of computer systems

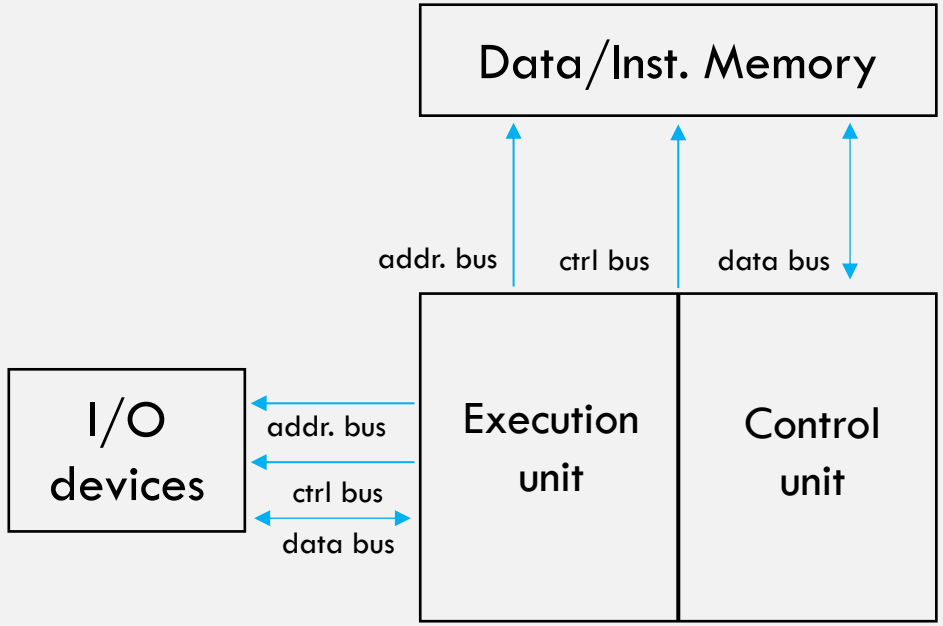
1.3 Architecture of computer

1.4 Application of microprocessor and microcontroller

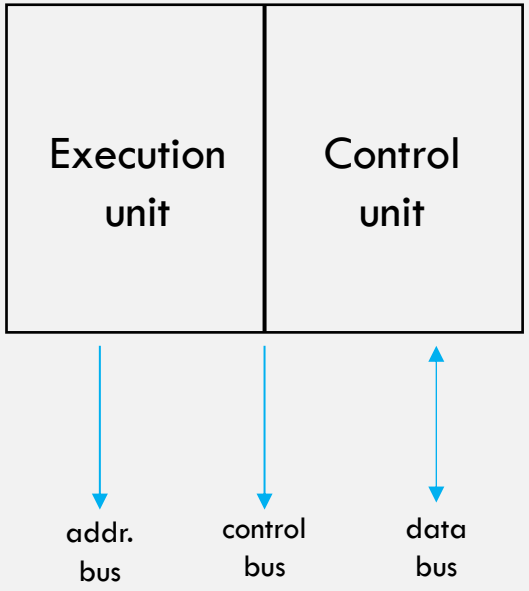
MICROPROCESSOR

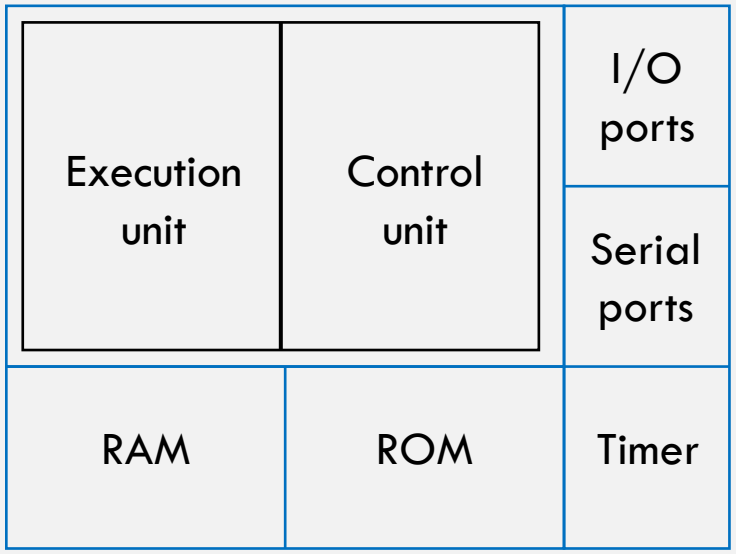
- An electronic device
- Programmable
- Reads binary instructions from memory
- Process data according to instructions
- Provides result to output devices

Organization of computer system



Microprocessor and Microcontroller

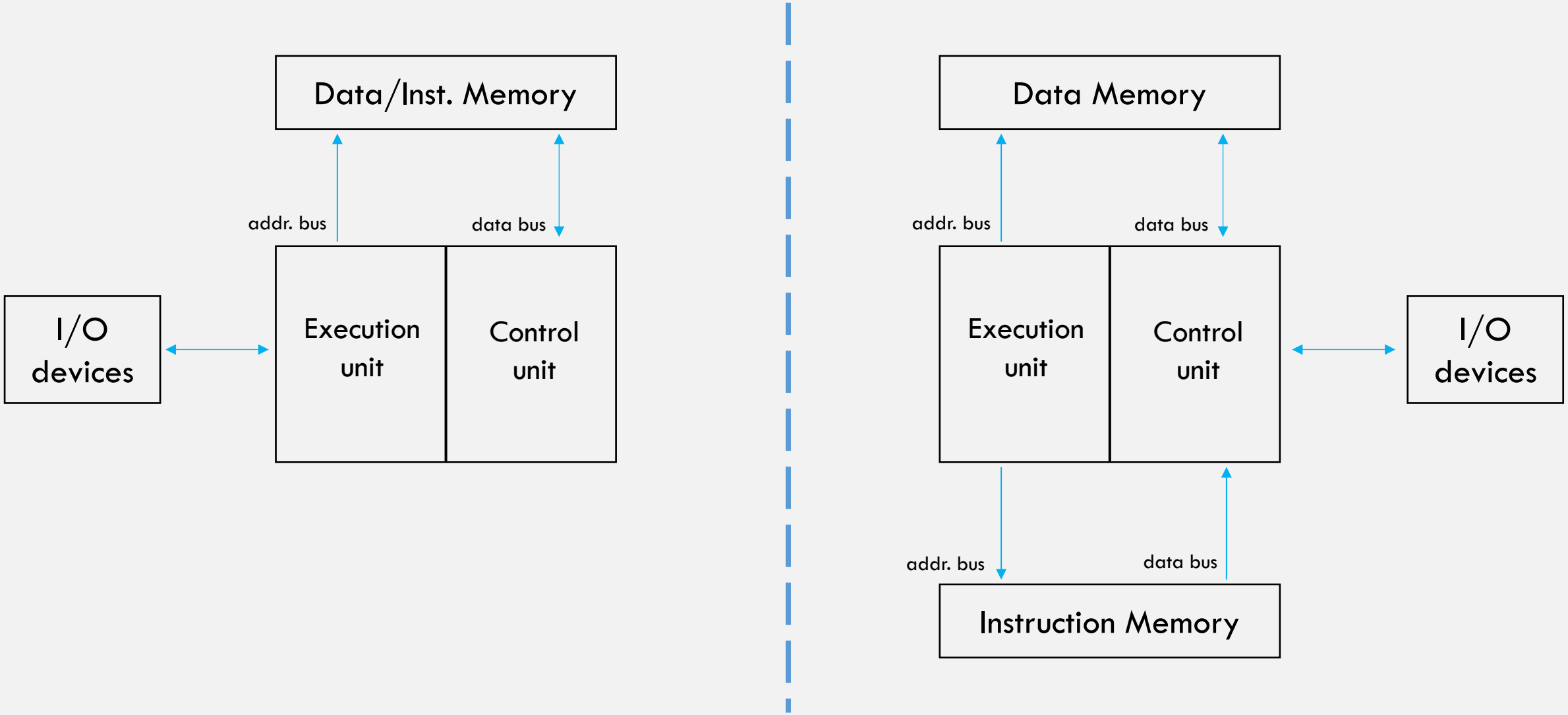




RISC vs CISC

Simpler instruction, hence simple instruction decoding	Complex instruction, hence complex instruction decoding
Instruction comes undersize of one word	Instructions are larger than one word size
Instruction takes a single clock cycle	Instruction may takes more than a single cycle
More general purpose registers	Less general purpose registers
Simple addressing modes	Complex addressing modes
Heavily pipelined	Less pipelined

Von-neumann and Harvard Architecture



Application of microprocessor and microcontroller

Microprocessor

- Central Processing Unit (CPU) for computational tasks.
- General-purpose, requires external memory & peripherals.
- Powers devices needing high performance & flexibility.

Application Area

- Personal Computers and Laptops
- Servers and Mainframes
- High end Gaming Consoles
- Aerospace and Defense
- Automotive Industry
- Medical Imaging and Diagnostics

Application of microprocessor and microcontroller

Microcontroller

- Self-contained SoC (System on Chip):
- Integrates CPU, memory, and I/O peripherals.
- Designed for embedded systems – optimized for specific tasks.
- Low power, cost-effective, and compact vs. microprocessors.

Application Area

- Consumer Electronics
- Home automation
- Automotive Systems
- Medical Devices
- Industrial Automation
- IoT devices